ICSC Competition Solutions – Rubaika Zaidi

Problem A:

Solution:

1) **1936:** Alan Turing proposes the Universal Turing Machine (foundation of modern computer science)
2) **1947:** Invention of the transistor at Bell Labs by John Bardeen, Walter Brattain, and William Shockley
3) **1958:** Invention of the integrated circuit (IC) by Jack Kilby (also independently developed by Robert Noyce)
4) **1969:** ARPANET goes live, the precursor to the modern Internet
5) **1981:** IBM Personal Computer (IBM PC) released, making personal computing mainstream
6) **1991:** The World Wide Web becomes publically avaliable, introduced by Tim Berners-Lee
7) **2000:** Dot-com era / Internet boom (widespread broadband Internet adoption)
8) **2007:** Apple introduces the iPhone, ushering the smartphone era
9) **2012:** Deep learning breakthrough (AlexNet wins ImageNet), marking the modern AI revolution
10) **2022:** Public release of ChatGPT / Generative AI becomes mainstream

Problem B:

Solution (Java):

```
public static int[][] generate_shape(int n, String shape) {


    int[][] grid = new int[n][n];


    if (shape.equals("checkerboard")) {


        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                grid[i][j] = (i + j) % 2;
```

```
            }
        }

    } else if (shape.equals("diamond")) {

        int center = n / 2;

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (Math.abs(i - center) + Math.abs(j - center) <= center) {
                    grid[i][j] = 1;
                } else {
                    grid[i][j] = 0;
                }
            }
        }

    }

    return grid;
}
```

Problem C:

Solution:

a) **Predict the transistor count in 2026**

Given:

- Initial transistor count (T_0) = 2300
- Reference year = 1971
- Target year = 2026
- Time elapsed (t) = 2026 − 1971 = 55 years

Formula:

$T = T_0 \times 2^{(t/2)}$

Substituting the values:

$T = 2300 \times 2^{(55/2)}$

$T = 2300 \times 2^{27.5}$

$2^{27.5} \approx 189,812,531$

$T = 2300 \times 189,812,531$

$T \approx 436,568,821,300$

Answer: The predicted transistor count in 2026 is approximately 436.6 billion transistors.

b) Compare with an actual 2026 processor

A modern processor such as the Apple M5 is estimated to have around 40 billion transistors.

Comparison:

- Predicted by Moore's Law: 436.6 billion transistors
- Estimated Apple M5: 40 billion transistors

Explanation:

The prediction is much higher than the estimated transistor count of the Apple M5. Moore's Law accurately described transistor growth for many decades, but in recent years the growth has slowed because of physical limitations, manufacturing costs, heat generation, and power consumption. Modern processors also focus on improving performance and efficiency rather than simply increasing the number of transistors.

c) Find the best fitting doubling-period

Formula:

$T = T_0 \times 2^{(t/d)}$

where d is the doubling period.

Substituting the values:

$40,000,000,000 = 2300 \times 2^{(55/d)}$

Rearranging the formula:

$d = 55 \div \log_2(40,000,000,000 \div 2300)$

$d = 55 \div 24.05$

$d \approx 2.3$ years

Answer: The best-fitting doubling period is approximately 2.3 years.

Conclusion: The transistor count has continued to grow exponentially over time, but instead of doubling exactly every 2 years, the data suggests it now doubles approximately every 2.3 years. This shows that Moore's Law still provides a good estimate, although the rate of growth has slowed slightly.

Problem D:

Solution(Java):

```
public static int[] processGame(int[][] events, int H) {


    // Stable sort by frame number

    Arrays.sort(events, Comparator.comparingInt(a -> a[1]));


    int hp1 = H;

    int hp2 = H;


    int i = 0;


    while (i < events.length) {
```

```
    int currentFrame = events[i][1];

    // Process every event in the current frame
    while (i < events.length && events[i][1] == currentFrame) {

        int player = events[i][0];
        int attack = events[i][2];

        if (player == 1) {
            hp2 -= attack;
        } else {
            hp1 -= attack;
        }

        i++;
    }

    // Clamp HP to minimum 0
    hp1 = Math.max(0, hp1);
    hp2 = Math.max(0, hp2);

    // Check for KO only after all events in this frame
    if (hp1 == 0 || hp2 == 0) {
        break;
    }
}
```

```
    return new int[]{hp1, hp2};
}
```

Problem E:

Solution:

A) PageRank measures the importance of a web page based on the number and quality of links pointing to it. It uses the random surfer model, where a user randomly clicks links on web pages.

The damping factor d=0.85 represents the probability that the surfer follows a link (85%). With probability 1−d=0.15, the surfer jumps to a random page. This prevents the algorithm from getting stuck in loops or dead ends.

The PageRank equation is recursive because the rank of a page depends on the ranks of the pages linking to it. Since every page depends on every other page, the values cannot be calculated directly.

In practice, PageRank is solved using iterative computation. All pages are initially assigned equal rank, and the formula is repeatedly applied until the PageRank values change by only a very small amount (they converge).

B)

Let:

- N = total number of pages
- a = number of AI pages
- α = a/N = fraction of AI pages
- SA = total PageRank of all AI pages
- SH = total PageRank of all human pages

Since the total PageRank of all pages is equal to 1,

$$SA + SH = 1$$

Therefore,

$$SH = 1 - SA$$

Every page receives a random jump contribution equal to (1 − d)/N. Since there are a = αN AI pages, the total random jump contribution received by the AI pages is:

$$\alpha N \times (1 - d)/N = \alpha(1 - d)$$

Now consider the contribution from hyperlinks.

Human pages distribute their PageRank uniformly among all pages. Therefore, the fraction of PageRank flowing from human pages to AI pages is α. Hence, the PageRank contribution from human pages is:

$$d\alpha SH$$

AI pages only link to other AI pages. Therefore, all of the PageRank already present in the AI pages remains within the AI group. Hence, the PageRank contribution from AI pages is:

$$dSA$$

Adding all three contributions, we obtain:

$$SA = \alpha(1 - d) + d\alpha SH + dSA$$

Substituting $SH = 1 - SA$,

$$SA = \alpha(1 - d) + d\alpha(1 - SA) + dSA$$

Expanding the brackets,

$$SA = \alpha(1 - d) + d\alpha - d\alpha SA + dSA$$

Since $\alpha(1 - d) + d\alpha = \alpha$, the equation simplifies to:

$$SA = \alpha - d\alpha SA + dSA$$

Moving all terms containing SA to the left-hand side,

$$SA + d\alpha SA - dSA = \alpha$$

Factoring out SA,

$$SA(1 - d + d\alpha) = \alpha$$

Finally, dividing both sides by $(1 - d + d\alpha)$ gives:

$$SA = \alpha / (1 - d + d\alpha)$$

Hence, the required closed-form expression for the total PageRank of all AI pages is:

$$\text{ANSWER: } SA = \alpha / (1 - d + d\alpha)$$

C)

The tipping point occurs when the total PageRank of AI pages is equal to the total PageRank of human pages.

Therefore,

$$SA = SH = 1/2$$

From part (b),

$$SA = \alpha / (1 - d + d\alpha)$$

Substituting SA = 1/2,

$$1/2 = \alpha / (1 - d + d\alpha)$$

Cross-multiplying,

$$1 - d + d\alpha = 2\alpha$$

Rearranging the terms,

$$1 - d = 2\alpha - d\alpha$$

Factoring out α,

$$1 - d = \alpha(2 - d)$$

Dividing both sides by (2 − d),

$$\alpha = (1 - d) / (2 - d)$$

Substituting d = 0.85,

$$\alpha = (1 - 0.85) / (2 - 0.85)$$

$$\alpha = 0.15 / 1.15$$

$$\alpha \approx 0.1304$$

Therefore,

$$\textit{α ≈ 0.13, or approximately 13\%.}$$

Conclusion: The tipping fraction is approximately 13%. This means that if AI-generated pages make up about 13% of all web pages, they will collectively r eceive 50% of the total PageRank in this model. This occurs because AI pages only link to other AI pages, causing PageRank to circulate within the AI network and increasing their collective importance.